

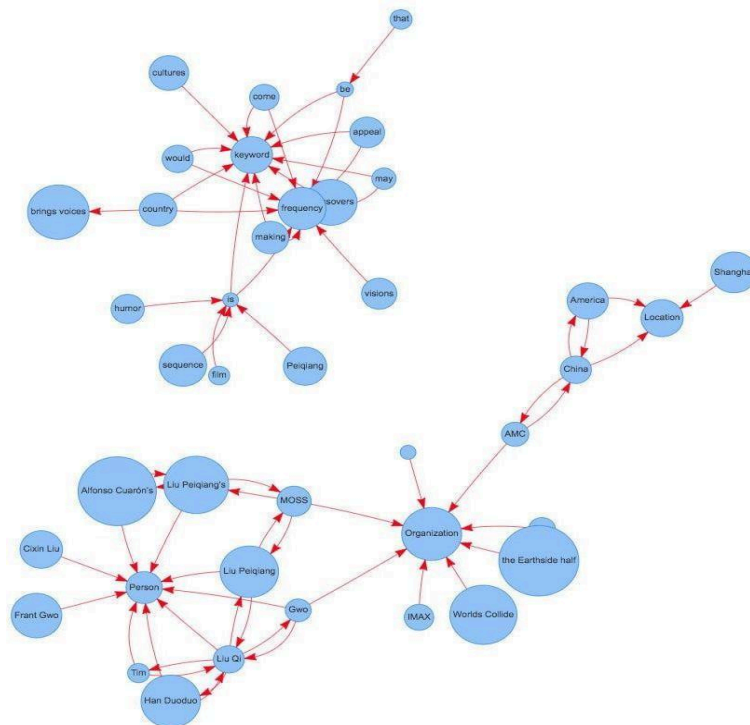
 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: TextRank for Document Summarization	
Experiment No: 10	Date:	Enrolment No: 92410133034

Aim: TextRank for Document Summarization

IDE: Google Colab

Theory:

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.



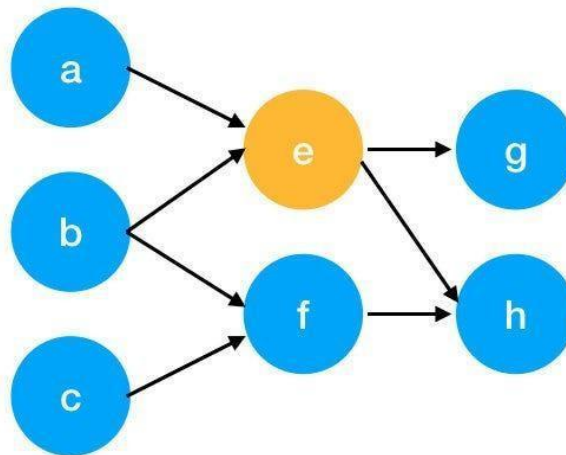
Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: TextRank for Document Summarization	
Experiment No: 10	Date:	Enrolment No: 92410133034

$$S(V_i) = (1 - d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- $S(V_i)$ - the weight of webpage i
- d - damping factor, in case of no outgoing links
- $In(V_i)$ - inbound links of i, which is a set
- $Out(V_j)$ - outgoing links of j, which is a set
- $|Out(V_j)|$ - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$\begin{aligned}
 In(v_e) &= \{a, b\}, j \in \{a, b\} \\
 \sum_{j \in \{a, b\}} \frac{1}{|Out(V_j)|} S(V_j) &= \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b) \\
 &= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b) \\
 &= S(V_a) + \frac{1}{2} S(V_b)
 \end{aligned}$$

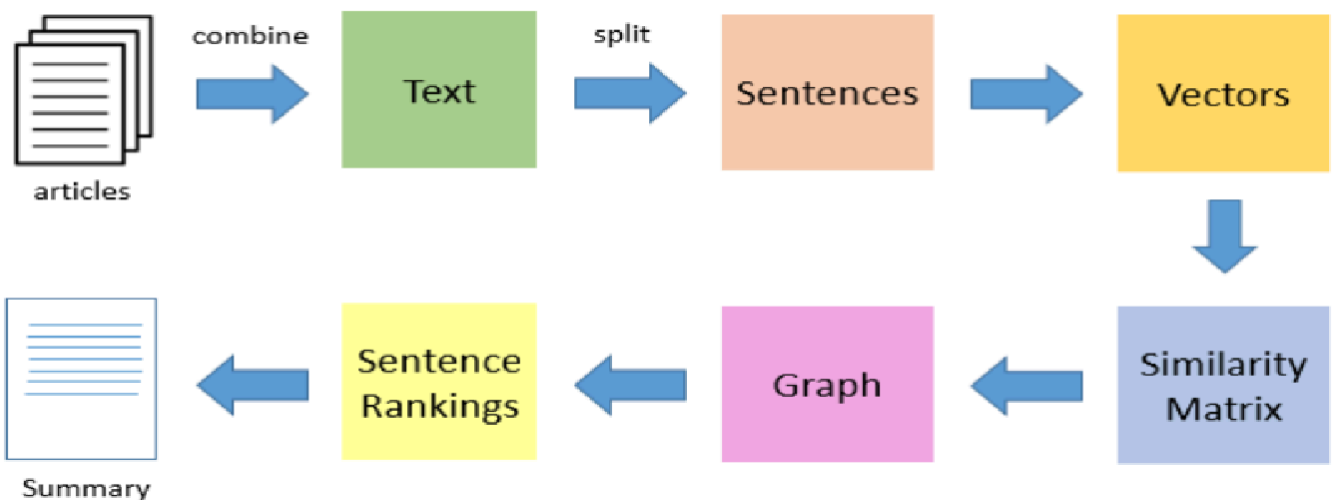
We can get the weight of webpage e by the following function.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: TextRank for Document Summarization	
Experiment No: 10	Date:	Enrolment No: 92410133034

$$S(V_e) = (1 - d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

TextRank for document Summarization



TextRank works in the following steps:

1. Tokenize documents into sentences.
2. Preprocess each sentence in the document.
3. Count key phrases and normalize them or produce TFIDF Matrix, you can also use any kind of vectorization such as spacy vectors.
4. Calculate the Jaccard Similarity between sentences and key phrases.
5. Rank the sentences with higher significance.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: TextRank for Document Summarization	
Experiment No: 10	Date:	Enrolment No: 92410133034

Pre Lab Exercise:

1. How keyword extraction is similar to document summarization process?
Both use TextRank and graph-based ranking to identify important parts of text. They analyze relationships between words/sentences to find important information.
2. How keyword extraction is different to document summarization process?
Keyword extraction selects important words or phrases, while summarization selects important sentences to represent the whole document.
3. Limitation of TextRank as a summarizer
TextRank does not understand meaning or context, may miss important sentences, and can include less relevant sentences in summary.

Program (Code):

```
[9]
✓ 9s !pip install pytextrank spacy

import spacy
import pytextrank

# Load model
nlp = spacy.load("en_core_web_sm")
nlp.add_pipe("textrank")

# Sample text
text = """Natural language processing is a field of artificial intelligence.
It helps computers understand human language.
It is widely used in chatbots, translation and speech recognition.
Many companies use NLP for automation and analysis."""

doc = nlp(text)

# Extract summary sentences
for sent in doc._.textrank.summary(limit_phrases=5, limit_sentences=2):
    print(sent)
```

Natural language processing is a field of artificial intelligence.
It helps computers understand human language.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: TextRank for Document Summarization	
Experiment No: 10	Date:	Enrolment No: 92410133034

Observation:

TextRank selects important sentences based on their relationships and ranks them to form a summary.

Post Lab Exercise:

1. Write about “your ownself” (in not more than 500 words□ You know better about you, rather than ChatGPT!!). Generate the summary of your portfolio in
 - a. 50% of the size of your portfolio
 - b. 25% of the size of your portfolio

Comment over the summary obtained. Which sentence you think should be there in your summary, but was not spotted by textrank? Rate the overall summary in the scale of 1-5 (with 1 as least and 5 as highest rating). Paste the code, your portfolio, output and your analysis.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: TextRank for Document Summarization	
Experiment No: 10	Date:	Enrolment No: 92410133034

```
[10]
✓ Os ▶ portfolio = ""I am a student of Information and Communication Technology.
I am interested in cybersecurity, networking and machine learning.
I have completed CCNA and worked on network topology and AI projects.
I like learning new technologies and solving real-world problems.""

doc = nlp(portfolio)

# 50% summary
print("50% Summary:")
for sent in doc._.textrank.summary(limit_sentences=2):
    print(sent)

# 25% summary
print("\n25% Summary:")
for sent in doc._.textrank.summary(limit_sentences=1):
    print(sent)
```

▼ ... 50% Summary:
I have completed CCNA and worked on network topology and AI projects.

I am interested in cybersecurity, networking and machine learning.

25% Summary:
I have completed CCNA and worked on network topology and AI projects.

Analysis

The summary captures important sentences but may miss some key details. Some important points may not be included due to lack of semantic understanding.

Overall summary rating: 3 / 5